VISVESVARAYA TECHNOLOGICAL UNIVERSITY

OFFICIAL COURSE SYLLABUS FRAMEWORK & TEMPLATE

(Outcome Based Education [OBE] & Choice Based Credit System [CBCS] Compliant)

SECTION 1: COURSE METRICS & REQUISITES

This framework defines the standard structure for all syllabus design and course delivery within the institution. Every course must be mapped against specific credit parameters and baseline requirements prior to delivery.

Course Code: XX21XX41 **Academic Year: 2026 - 2027**

Course Title: Data Structures and Algorithms **Semester: 4th (Even)**

Total Contact Hours: 40 Hours Lecture, 20 Hours Lab **Overall Valuation: 100 Marks**

Core Distribution: 3 Credits Lecture, 1 Credit Lab **Type: Integrated (Theory + Lab)**

1.1 Pre-requisite Mapping

Before registering for this course, students must have successfully cleared or demonstrated equivalent technical proficiency in the following foundational domains:

- **CS21PL15 - Problem Solving using C Programming: Mastery over memory management, pointers, structure allocations, arrays, and basic file operations.**
- **MA21BS21 - Discrete Mathematical Structures: Understanding of graph theory foundations, tree logic systems, set operations, permutations, and combinations.**

SECTION 2: COURSE OUTCOMES (CO) & MAPPING CODES

2.1 Course Outcomes (CO) Definitions

Upon successful completion of this course, students will be able to demonstrate measurable competency across five core dimensions:

- **CO-1 (Cognitive Level: L2 - Understand): Explain the operational mechanics, structural memory allocation layout, and architectural constraints of linear and non-linear data structures.**
- **CO-2 (Cognitive Level: L3 - Apply): Apply programmatic stack, queue, and linked-list logic flows to resolve complex real-world data processing challenges.**

- **CO-3 (Cognitive Level: L4 - Analyze): Analyze and calculate the time and space complexity variances of diverse sorting and searching algorithms using Big-O asymptotic notation.**
- **CO-4 (Cognitive Level: L3 - Apply): Implement and traverse advanced non-linear data topologies, including binary search trees, AVL configurations, and directed/undirected graphs.**
- **CO-5 (Cognitive Level: L5 - Evaluate): Evaluate an open-ended computational problem statement and determine the optimal data structure configuration to maximize algorithm execution speed and minimize memory consumption.**

SECTION 3: MODULAR SYLLABUS CONTENT BREAKDOWN

The 5-Module Syllabus Sequence

1.Module 1: Introduction, Complexity, and Linear Formations:Instructional Window: Weeks 1 to 3 (8 Hours).

Review of Dynamic Memory Allocation and pointer arithmetic. Definition and abstract data types (ADTs). Introduction to Asymptotic Notation: Big-O, Omega, and Theta metrics. Linear Data Structures: Static and Dynamic array tracking, sequential storage modeling, implementation of Single, Double, and Circular Linked Lists.

2.Module 2: Restricted Linear Structures - Stacks and Queues:Instructional Window: Weeks 4 to 6 (8 Hours).

The Stack ADT: Array and Linked-List representation models. Operations: Push, Pop, Peek. Applications of Stacks: Infix to Postfix expression conversion, Postfix evaluation matrix, recursion simulation. The Queue ADT: Linear Queue layout, Circular Queue logic, Double-Ended Queues (Deque), Priority Queue scheduling mechanics.

3.Module 3: Hierarchical Structures - Non-Linear Trees:Instructional Window: Weeks 7 to 9 (8 Hours).

Foundational Tree Terminology. Binary Trees: Properties, Array and Linked representation structures. Binary Tree Traversals: Pre-order, In-order, Post-order execution pathways. Binary Search Trees (BST): Insertion, Deletion, and Searching operations. Introduction to self-balancing systems: AVL Tree balance factors and basic rotational mechanics.

4.Module 4: Interconnected Topologies - Graphs and Hashing:Instructional Window: Weeks 10 to 12 (8 Hours).

Graph Foundations: Directed vs. Undirected frameworks, degree patterns. Representation Formats: Adjacency Matrix and Adjacency List. Graph Traversal Methodologies: Breadth-First Search (BFS) and Depth-First Search (DFS) logical tracks. Hashing: Hash functions, collision resolution mechanisms (Open Addressing vs. Chaining structures).

5.Module 5: Algorithmic Paradigms - Sorting and Searching:Instructional Window: Weeks 13 to 15 (8 Hours).

Internal vs. External sorting models. Advanced Sorting: Divide-and-Conquer paradigm, design and analysis of Merge Sort and Quick Sort algorithms. Searching Strategies: Linear Search vs. Binary Search execution constraints. Heap Structure: Max-Heap and Min-Heap formations, execution of Heap Sort.

SECTION 4: INTEGRATED LABORATORY PRACTICAL MODULES

Every student must independently construct, compile, and execute the following laboratory programs within a proctored environment to satisfy the practical requirements of the course:

1. **Dynamic Memory Array Simulator: Design a C/C++ program to maintain a dynamic array of student record structures, implementing auto-scaling memory allocation logic when the storage boundary is crossed.**
2. **Expression Parser Engine: Develop a stack-based software parser that takes a raw string configuration, converts it from an Infix expression to a valid Postfix token sequence, and calculates the final balance value.**
3. **Real-Time Queue Scheduler: Build a functional simulation of an operating system CPU process scheduler utilizing a multi-level Priority Queue structural model.**
4. **BST Search Indexer: Construct an index-tracking system for a text file database using a Binary Search Tree (BST) map. Provide functionalities to track occurrences and calculate the depth of any chosen word.**

SECTION 5: RECOMMENDED LEARNING RESOURCES

5.1 Primary Prescribed Textbooks (Latest Editions)

1. **Ellis Horowitz, Sartaj Sahni, and Susan Anderson-Freed: *Fundamentals of Data Structures in C*, Silicon Press / Universities Press.**
2. **Mark Allen Weiss: *Data Structures and Algorithm Analysis in C/C*++, Pearson Education.**

5.2 Reference Material and Digital Library Subscriptions

1. **Reema Thareja: *Data Structures Using C*, Oxford University Press.**
2. **Seymour Lipschutz: *Data Structures (Schaum's Outline Series)*, McGraw Hill Education.**
3. **NPTEL / SWAYAM Central Portal: *Programming, Data Structures, and Algorithms using Python/C* – Certified Course Series by IIT Madras.**